Placement progress and notes

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1 Monthly Progress

Month 1: 23/02/2026 - 20/03/2026

Overview

Spent the first week familiarising myself with beam optics concepts and Xsuite functions. Then began understanding and editing a code for a compact ring with 90° phase advance.

Work completed

- Initially read papers to understand how the ring functions and what processes go into it.
- Made notes on initial ring design code and started compiling work into a GitHub repository.
- Added sextupoles to the ring design and performed chromaticity correction.
- Corrected beta-functions to create uniform Twiss plot.
- Corrected working point and found two potential candidates.

Problems encountered

Breaking the ring: `Invalid n1`, `Invalid n2`, and `Invalid n3` errors arose when trying to add sextupoles. This scenario halted progress the most.

- **Solution:** There was no set solution to this error but there were some common causes: changing the geometry of the ring, not matching/rematching the quadrupole strengths. The 'breakage' could be identified by computing a Twiss with specified conditions.

Sextupole positioning: This would cause ring breakage. It took a week or so to identify sextupole positions that worked.

- **Solution:** I tried creating a function that identified the best positions for sextupoles, but the best solution was trial and error and knowledge of ring properties.

Large beta-functions: Changing parameters resulted in increased beta-functions

- **Solution:** Consistently running matching functions and identifying best positions and working points.

Results and progress

- The initial compact ring now has sextupoles inserted: there are currently 16 in each arc (8 focusing, 8 defocusing).
- Strengths and beta-functions have been corrected
- Two potential working points have been identified and plotted on resonance graphs to show suitability

- Using these working points resulted in reduced beta-functions, more uniform Twiss plots, and slightly reduced sextupole strengths.
- Another configuration with a decreased triplet drift length is being explored and shows signs of reduced beta-functions
- Chromatic functions are plotted and show slightly different, but periodic, results for both working points.

Next steps

- Further refine suitability of the working point
- Track off-momentum particles
- Track dynamic aperture and momentum acceptance
- Work on different sextupole configurations with different positions, families, and cell lengths in parallel

Skills developed

- **Knowledge of beam optics concepts:** Twiss parameters, chromaticity, tune, basics of synchrotron radiation, and dynamic aperture
- **Xsuite:** Building rings and inserting elements, calculating Twiss plots, creating matching functions

Month 2: 23/03/2026 - 24/04/2026

Overview

Completed dynamic aperture and momentum acceptance studies for different configurations. Experimented with slicing, integrators, and fringe fields. Now introducing misalignments and corrections.

Work completed

- Plotted DA/MA for 7 different sextupole configurations
- Optimised configurations to increase DA/MA
- Changed number of slices in integrators to increase quality of plots
- Introduced fringe fields for every magnet type and compared with "perfect" results
- Performed frequency map analysis
- Introduced misalignments, inserted BPMs and correctors, and performed orbit correction

Problems encountered

Reducing non-linearities: Chromatic correction is working for *dqx* and *dqy* but often plotting tune against momentum deviation showed that *ddqx* and *ddqy* had not been properly corrected. This contributed to smaller DA/MA. Performing chromaticity correction to get these values to 0 often failed as it was too much of a jump.

- **Solution:** The chromaticity correction tries to match to 80% of the value of *ddqx* and *ddqy*. If this is successful, it iterates through 80% of the resulting values until the code fails. It takes the strengths from the iteration with the smallest second order chromaticity.

Code organisation: Many scripts and results in different places

- **Solution:** Everything has been modularised so it is easier to see where errors arise and can edit things more efficiently. This will also help with later lattice designs.

Results and progress

- Increasing the number of slices for the quadrupoles created smoother curves in the DA/MA plots, but there were no differences when increasing the slices for bends and quadrupoles.
- Fringe fields reduced the less stable configurations' DA/MA considerably, but not as much for stable configurations.
- Added another focusing sextupole pair to the configuration with two sextupole pairs at 180° phase advance, which dramatically improved momentum acceptance.
- Frequency map analysis calculated and shows stable diffusion
- BPMs and correctors are currently placed around every quadrupole (horizontal and vertical) and the MICADO algorithm is used for corrections
- With corrections, the orbits are currently oscillating at less than 1mm

Next steps

- Studies on spin tracking and polarization build-up time
- Looking at integration with the rest of the injector complex using current macroparticle distribution
- Assess injection efficiency

Skills developed

- **Dynamic aperture and momentum acceptance:** how it is calculated and the effects of adjusting integrators and slices
- **Knowledge of beam optics concepts:** fringe fields, FMA, BPMs, correctors, synchrotron radiation damping

2 Notes and Concepts

Overview of processes within the ring

2.0.1 Injection - electron enters the ring

- An electron bunch is injected at a defined energy
- At this point:
  - Position is near the closed orbit
  - Momentum is close to design value
  - Spins are unpolarized
- The electron now starts circulating

2.0.2 First turn - Bending in Dipoles

- The electron enters a dipole magnet
- The magnetic field applies the Lorentz force: $\vec{F} = q\vec{v} \times \vec{B}$
- Electron trajectory curves and now travels along an arc
- This defines the bending radius: $\rho = \frac{P}{eB}$

2.0.3 Synchrotron radiation happens immediately

- Because the electron is accelerating (curved motion), it emits photons and loses energy.
- Energy loss per turn: $U_0 \propto \frac{E^4}{\rho}$

2.0.4 RF Cavities

- After losing energy, due to synchrotron radiation, in the dipoles:
  - The electron enters the RF cavity
  - The RF electric field accelerates it longitudinally
  - Energy is restored
- However, radiation reduces total momentum, whereas RF only restores longitudinal momentum
- This imbalance creates damping

2.0.5 Radiation Damping

- Over many turns, horizontal, vertical, and longitudinal oscillations shrink
- This occurs because electron radiation losses exceed the energy restored by the RF cavity (it doesn't restore the transverse momentum)
- Therefore, after a full turn, the total momentum is restored but the transverse components are reduced
- Transverse momentum follows $x_n = x_0 e^{-n/\tau_x}$ where $\tau_x = \frac{2E}{J_x U_0}$
- Vertical oscillations also increase path length and radiation slightly
- If a particle has a higher energy, it will radiate stronger, therefore losing more energy per turn
- The beam shrinks towards equilibrium

2.0.6 Quantum Excitation

- Radiation is not continuous
- Electrons emit discrete photons randomly
- Each photon changes energy slightly and introduces noise
- Damping shrinks motion and quantum fluctuations increase motion
- Eventually, damping = excitation and the beam reaches equilibrium emittance and energy spread

2.0.7 Spin Begins Evolving

- Electron spins precess as it moves in the magnetic field according to the Thomas-BMT equation
- Spin tune: $\nu_s = a\gamma$ spin rotates many times per revolution
- Initially spins are randomly

2.0.8 Sokolov-Ternov Effect Starts

2.0.9 Polarization Build-Up

2.0.10 Equilibrium State

- Fixed emittance
- Fixed energy spread
- Stable bunch length
- Stable polarization

Glossary

Alpha Function A Twiss parameter that describes the slope of the beta function and the focusing/defocusing of the beam envelope. It is defined as,

$$\alpha(s) = -\frac{1}{2}\frac{d\beta}{ds}$$

and determines the correlation between transverse position and angle..

Analysing Power The sensitivity of a polarization measurement to the spin orientation of the beam. It quantifies how strongly an observable depends on the polarization direction and sets the statistical precision of polarization measurements..

Beam Position Monitors .

Beam Rigidity Quantifies how difficult it is to bend a charged particle beam using magnetic fields. It is defined by

$$B_\rho = \frac{p}{q}$$

where p is the particle momentum and q is its charge, and it sets the required dipole field strength for a given curvature..

Beta Function A Twiss parameter that describes how the amplitude of betatron oscillations varies along the lattice. The rms beam size at a given location is described by,

$$\sigma_x(s) = \sqrt{\epsilon\beta(s)}$$

showing its direct connection to emittance and beam envelope..

Betatron Motion Referes to the transverse oscillatory motion of particles around the reference orbit caused by alternating-gradient focusing. This motion is stable when the lattice satisfies periodic focusing conditions..

Betatron Tune The number of transverse oscillations a particle makes in one full revolution of the ring. It is defined as

$$Q = \frac{1}{2\pi}\oint \frac{ds}{\beta(s)}$$

and must be chosen to avoid resonance conditions..

Chromaticity Describes the depencdence of the betatron tune on particle momentum. It arises because quadrupole focusing strength depends on momentum and is quantified by

$$Q' = \frac{dQ}{d(\Delta p/p)} = -\frac{1}{4\pi}\oint k(s)\beta(s)ds$$

.

Depolarization Any mechanism that reduces the net polarization of a beam over time. Depolarization can arise from spin-orbit coupling, field errors, energy spread, and stochastic effects from synchrotron radiation..

Dipole Magnet A dipole magnet produces a uniform transverse magnetic field used to bend particle trajectories along the reference orbit. The total integrated dipole field around the ring must produce an angle of 2π..

Dispersion Function Describes how the transverse position of a particle depends on its relative momentum deviation. It appears in the relation

$$x(s) = x_\beta(s) + D(s)\frac{\Delta p}{p}$$

and serves as a fundamental measure of beam quality..

Dynamic Aperture The maximum stable area within an accelerator where particles can circulate without being lost due to nonlinear magnetic effects..

Emittance An invariant of linear transverse motion representing the area occupied by the beam in phase space. It is defined by

$$\epsilon = \gamma x^2 + 2\alpha x x' + \beta x'^2$$

and serves as a fundamental measure of beam quality..

Equilibrium Polarization The steady-state polarization level reached when polarization build-up matches depolarization mechanisms. In electron storage rings, it is determined by the competition between the Sokolov-Ternov effect and spin diffusion..

Focusing Function Characterises the net transverse focusing strength along the lattice. It is given by

$$K(s) = k(s) + \frac{1}{\rho^2(s)}$$

including contributions from quadrupoles and weak focusing from dipole curvature..

FODO Lattice A periodic focusing structure consisting of alternating focusing and defocusing quadrupole magnets separated by drift spaces and dipoles. It is widely used in synchrotrons due to its simplicity and stable optical properties..

Frequency Map Analysis .

Fringe Fields .

Gamma Function A Twiss parameter associated with beam divergence and angular spread. It is defined as

$$\gamma(s) = \frac{1 + \alpha^2(s)}{\beta(s)}$$

.

Harmonic Number The ratio of the RF cavity of the beam. It determines the number of RF buckets around the ring and is defined as

$$h = \frac{f_{RF}}{f_{rev}}$$

.

Hill's Equation The linear differential equation governing transverse particle motion in a synchrotron. Its written as

$$x''(s) + K(s)x(s) = 0$$

and its periodic solutions describe betatron oscillations..

Invariant Spin Field A vector field defined along the closed orbit that specifies the stable spin direction at each point in phase space. It provides a framework for understanding spin stability and depolarization in storage rings..

Linear Approximation The linear approximation assumes that magnetic fields can be expanded to first order transverse displacement. Under this assumption, only dipole and quadrupole field components are retained, leading to linear equations of motion..

Momentum Acceptance The range of energy deviations a particle can have relative to the design energy while remaining stable..

Momentum Compaction Factor Describes how the closed-orbit length changes with particle momentum. It is defined as,

$$\gamma_p = \frac{\Delta L/L}{\Delta p/p}$$

and plays a key role in longitudinal beam dynamics..

One-Turn Matrix The product of all transfer matrices around a full revolution of the ring. Its eigenvalues determine the betatron tune and the stability of transverse motion..

Phase Advance The accumulated betatron phase between two points in the lattice. It is given by

$$\psi(s) = \int_0^s \frac{ds'}{\beta(s')}$$

and its total over one turn defines the tune..

Phase Focusing The RF mechanism that restores particles towards a stable synchronous phase. It provides longitudinal stability by correcting early or late particle arrival times..

Polarization The degree to which particle spins are aligned along a preferred direction. It is commonly defined as

$$P = \frac{N_\uparrow - N_\downarrow}{N_\uparrow + N_\downarrow}$$

where $N_\uparrow$ and $N_\downarrow$ are the numbers of particles with spin aligned or anti-aligned with the reference axis..

Polarization Build-Up Time The characteristic time constant describing how quickly polarization grows toward its equilibrium value. For the Sokolov-Ternov effect, it appears in the exponential growth law,

$$P(t) = P_\infty(1 - e^{t/\tau_P})$$

.

Polarization Wiggler A magnetic insertion device designed to enhance synchrotron radiation damping, therefore reducing polarization build-up time. Wigglers increase spin-flip rates while minimally perturbing the overall lattice optics when properly designed..

Quadrupole Magnet A quadrupole magnet produces magnetic field gradients that provide linear transverse focusing. The field components satisfy $B_x = gy$ and $B_y = gx$ leading to focusing in one plane and defocusing in the other. The constant g is the gradient of the magnetic field and characterises the focusing strength of the quadrupole lens in both transverse planes..

Reference Orbit The ideal closed trajectory followed by a particle with nominal momentum and no transverse displacement. All transverse coordinates and deviations are defined relative to this orbit..

Resonant Depolarization A method used to determine beam energy with extreme precision by using an external oscillating magnetic field to flip the spins of particles when the field frequency matches their natural precession frequency..

RF Cavity A resonant structure that provides oscillating electric fields for particle accelerations and longitudinal focusing. It defines the stable phase and energy of the synchronous particle..

Sextupole Magnet A non-linear magnetic element used primarily to correct chromaticity. Sextupoles are placed in regions with high dispersion to compensate momentum-dependent focusing errors..

Slip Factor Determines how the revolution period of a particle changes with momentum. It is defined as,

$$\eta = \frac{1}{\gamma^2} - \alpha_p$$

and governs the longitudinal stability regimes..

Sokolov-Ternov Effect .

Spin Diffusion .

Spin Precession .

Spin Resonance .

Spin Tune .

Spin-Orbit Coupling .

Stability Condition The stability condition for transverse motion requires that the trace of the one-turn matrix satisfy $|Tr(M)| < 2$. This ensures bounded betatron oscillations..

Symplectic Integrators .

Synchrotron A circular accelerator in which magnetic fields and RF frequency are synchronised with particle energy. It enables acceleration and storage of high-energy particle beams..

Synchrotron Oscillation Longitudinal oscillations of particles in energy-phase space around the synchronous particle. They occur at much lower frequencies than betatron oscillations..

Synchrotron Radiation .

Thomas-BMT Equation .

Transfer Matrix A linear operator that maps particle coordinates through an accelerator element. Successive multiplication of matrices describes transport through the full lattice..

Transition Energy The beam energy at which the slip factor becomes zero. At this point, the character of longitudinal motion changes, making operation challenging..

Transverse Coordinates Describe particle motion perpendicular to the reference orbit are given by (x, x', y, y'). They represent position and angle in the horizontal and vertical planes..

Twiss Parameters The Twiss parameters $\beta(s)$, $\alpha(s)$, and $\gamma(s)$ describe the evolution of the beam envelope in a linear lattice. They provide a complete description of transverse linear optics..

Vertical Polarization .

Working Point .